THESIS B — RESEARCH CONCLUSIONS

Tidal Triggering of Great Earthquakes

The Syzygy Question: A Pre-Registered Test

NULL RESULT

Tidal triggering is not supported

Study period:	1960–2026 (USGS instrumental era)
Catalog:	USGS FDSNWS — 902 M7.0+ earthquakes retrieved
Primary test:	M7.5+ subduction zone megathrusts, syzygy ±3 days
Significance level:	$\alpha = 0.01$ (pre-registered, stricter than conventional 0.05)
Key result:	$p = 0.9175$, RR = 0.871× — no evidence of tidal triggering
Ide et al. 2016:	Original signal not replicated. Effect appears to be a statistical artifact of small N and post-hoc threshold selection.

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1. Executive Summary

> **Finding:** Do great earthquakes preferentially occur near the time of new or full moon? After rigorous analysis of 65 years of global seismicity, the answer is **no**. The rate of M7.5+ subduction zone earthquakes during syzygy windows is **below** the null expectation (RR = 0.871×, p = 0.9175). Tidal triggering of great earthquakes is not supported.

This document reports the complete results of Thesis B of the Earth Systems Observatory, a pre-registered investigation into whether the fortnightly tidal stress cycle statistically elevates the occurrence rate of great subduction zone earthquakes. The test was motivated by Ide, Yabe, and Tanaka (2016, *Nature Geoscience*), who reported that M≥8.2 earthquakes preferentially nucleate near times of high tidal stress. That finding has been contested on methodological grounds, including small sample sizes, multiple testing, and possible post-hoc threshold selection.

Our analysis covers 902 M7.0+ earthquakes spanning 1960–2026, filtered to 536 subduction zone events, and declustered to 390 mainshocks using the Gardner-Knopoff algorithm. The primary pre-registered test — exact one-sided binomial test of M7.5+ events in syzygy windows — finds no enrichment at any conventional significance threshold. All secondary analyses (magnitude sweep, window sensitivity, robustness checks, temporal split, zone breakdown) confirm the null result with high consistency.

The three greatest recorded earthquakes — the 1960 Chilean M9.5, 1964 Alaskan M9.2, and 2004 Sumatra-Andaman M9.1 — all fell within syzygy windows, but this is fully consistent with null expectation: approximately 40% of all events are expected to fall in syzygy windows by chance alone, and the actual rate in our sample (35.4%) is **below** that baseline.

2. Background and Motivation

2.1 The Physics of Tidal Stress

The Moon and Sun exert gravitational forces that vary across Earth's diameter, producing differential (tidal) forces. These manifest as body tides in the solid Earth (amplitude ∼30 cm), ocean tides, and accompanying crustal stress perturbations. The tidal stress field has a fortnightly periodicity: peak stress occurs at syzygy (new and full moon, when Moon-Earth-Sun alignment maximizes tidal forcing) and minimum stress at quadrature (first and last quarter moon).

Theoretical estimates of tidal stress on fault planes range from approximately 1–5 kPa for body tides alone, rising to 10–20 kPa in coastal subduction zones where ocean loading adds to the solid-Earth signal. These magnitudes are 3–4 orders of magnitude smaller than typical earthquake stress drops (1–10 MPa), raising the fundamental question: how could such a small perturbation modulate the occurrence of the largest earthquakes?

Several mechanisms have been proposed. Rate-and-state friction theory (Dieterich 1994) predicts that even small periodic stresses can modulate seismicity rates if the background loading rate is slow and the fault is near failure: $\Delta r/r \approx \Delta\tau / (A\sigma_n)$, where A is a dimensionless friction parameter (~0.01). For faults near critically-stressed failure, the amplification could be meaningful. Additionally, some researchers propose that tidal stress might trigger slow-slip events or episodic tremor and slip (ETS), which in turn could reduce effective normal stress and encourage mainshock nucleation. The ocean-loading mechanism is most plausible specifically for shallow oceanic thrust faults — exactly the subduction megathrusts targeted in this study.

2.2 The Ide et al. 2016 Finding

Ide, Yabe, and Tanaka (2016) analyzed the relationship between tidal stress phase and earthquake magnitude for global seismicity. Their key claim was that the largest earthquakes (M≥8.2) cluster preferentially in the high-stress half of the tidal cycle, with the relationship strengthening monotonically with magnitude. They proposed a *cascade triggering* model in which small tidal perturbations initiate slow-slip that enables larger ruptures to nucleate.

The paper attracted significant criticism. Key concerns included:

- **Small sample size:** Only 9–12 earthquakes in the highest magnitude bins — far too few for robust statistics.
- **Multiple testing:** The tidal stress threshold and magnitude cutoff were not pre-registered, creating opportunities for inadvertent p-hacking.
- **Ambiguous tidal metric:** Ide et al. used tidal stress amplitude rather than phase, which conflates syzygy and quadrature cycles differently than a phase-based test.
- **Declustering:** Aftershock sequences were not systematically removed, which can artificially inflate any temporal clustering signal.

This study was designed to provide a rigorous, pre-registered replication test using a simpler and more transparent methodology: a direct comparison of earthquake occurrence rates during syzygy versus non-syzygy periods.

2.3 Pre-Registration and Falsification Criteria

The following parameters were committed before data collection:

Parameter	Value
Primary magnitude threshold	M7.5+ (subduction zone megathrusts only)
Syzygy window	±3 days of new or full moon (~40.6% of calendar time)
Statistical test	Exact one-sided binomial, $P(X \geq k \mid n, p_0 = 0.406)$
Significance level (α)	0.01 (stricter than 0.05, justified by prior publication bias toward positive results)
Declustering	Gardner-Knopoff algorithm (standard M-dependent windows)
Falsification criterion	$p \geq 0.01$ with $N \geq 20$ events → tidal triggering not supported

3. Data and Methods

3.1 Earthquake Catalog

Seismicity data were retrieved from the USGS Earthquake Hazards Program FDSNWS API, covering M7.0+ global events from 1960 January 1 through 2026 April 2. Data were downloaded in 5-year chunks to avoid API response size limits. A total of **902 events** were retrieved. Focal depths were not restricted; however, geographic filtering was applied to retain only events associated with the 19 major subduction zone systems (see Section 3.3).

The 1960 start date was chosen to correspond with the onset of reliable global seismic network coverage following the establishment of the World-Wide Standardized Seismograph Network (WWSSN). A temporal sensitivity test splitting the catalog at 1990 (when network coverage improved further with broadband instruments) confirms that results do not differ meaningfully between periods.

3.2 Lunar Phase Computation

Lunar phase angles were computed using the Meeus astronomical algorithm (Meeus 1998, *Astronomical Algorithms*, 2nd ed., Chapter 49). This deterministic calculation returns the elongation angle between the Moon and Sun as seen from Earth, ranging from 0° (new moon) through 90° (first quarter), 180° (full moon), 270° (last quarter), and back to 360°/0°. The algorithm accounts for the eccentricity of both the Earth-Moon and Earth-Sun orbits and achieves sub-degree accuracy, more than sufficient for classification into ±3-day syzygy windows.

Validation checks confirmed:

- 2024-01-11 (new moon): computed phase = 357.5° (expected ≈0°)
- 2024-01-25 (full moon): computed phase = 181.8° (expected ≈180°)

A syzygy window is defined as any period within ±3 days of a new or full moon (phase angle ≤ 36.7° or ≥ 323.3°, or 143.3° ≤ phase ≤ 216.7°). Under this definition, syzygy windows occupy approximately **40.6%** of calendar time — the null expectation rate $p_0 = 0.406$ used in all binomial tests.

3.3 Subduction Zone Filtering

Events were filtered to those falling within 50 km of 19 major subduction zone systems defined by published trench geometry. This retained 536 of 902 events (59.4%). The zone definitions cover all major convergent margins including the Cascadia, Alaska-Aleutian, Japan, Kuril-Kamchatka, Ryukyu, Izu-Bonin-Mariana, Philippine, New Hebrides, Tonga-Kermadec, Solomon Islands, New Britain, Sumatra-Java, Makran, Chile-Peru, Central America, and Caribbean systems.

3.4 Aftershock Declustering

The Gardner-Knopoff (1974) declustering algorithm was applied using magnitude-dependent time and distance windows. For each event, all subsequent events within the time-distance window are classified as aftershocks of the larger event. Three declustering layers were computed:

- **Standard (1.0×):** 536 subduction events → 390 mainshocks (146 aftershocks removed)
- **Extended (1.5× windows):** 536 events → 317 mainshocks (219 removed). Used as robustness check.

- **All settings (unconstrained):** 902 total events → 707 mainshocks. Used to test sensitivity to subduction zone restriction.

All primary results use the standard (1.0×) declustered catalog of 390 mainshocks.

3.5 Statistical Testing

The primary test is an **exact one-sided binomial test**: given n mainshocks of magnitude ≥ M, how many fall in syzygy windows? Under the null hypothesis, the probability of any event falling in a syzygy window is $p_0 = 0.406$. The one-sided p-value is $P(X \geq k \mid n, p_0)$. For the primary threshold (M7.5+, n=147), the exact binomial computation was used.

Secondary analyses included:

- **Permutation test:** 10,000 Monte Carlo shuffles, each simulating n events with independent probability p_0 of syzygy occurrence. Empirical p-value = proportion of simulations equaling or exceeding the observed count.
- **Clopper-Pearson confidence interval:** Exact 95% binomial CI on the observed syzygy rate.
- **Magnitude sweep (dose-response):** Binomial tests at M7.0+, M7.5+, M8.0+, M8.5+ thresholds. Bonferroni-corrected $\alpha = 0.01/4 = 0.0025$. Tests for monotonic increase in RR with magnitude (the Ide et al. prediction).
- **Window sensitivity:** ±1, ±2, ±3 day windows tested.
- **Robustness:** Extended 1.5× declustering; all tectonic settings.
- **Temporal split:** Pre-1990 vs post-1990.

4. Results

4.1 Primary Test

> **Primary result (pre-registered):** M7.5+ subduction earthquakes, syzygy ±3 days, N = 147 events. Syzygy hits: 52 (35.4%). Expected: 40.6%. Rate ratio: 0.871×. p = 0.9175. **Not significant** at α = 0.01.

The primary pre-registered test finds no evidence of tidal triggering. The observed syzygy rate (35.4%) is **below** the null expectation of 40.6%, with a rate ratio of 0.871× — meaning M7.5+ earthquakes are slightly, but non-significantly, *less* likely during syzygy windows than at other times. The one-sided p-value of 0.9175 is far above both conventional (0.05) and pre-registered (0.01) thresholds. The permutation test confirms this: p = 0.9127 across 10,000 shuffles.

To put this in practical terms: of 147 M7.5+ declustered mainshocks over 65 years, 52 fell in syzygy windows. Under a pure null hypothesis, we would expect approximately 40.6 × 147 ≈ 59.7 events. The observed count is **below** that expectation by 7.7 events, not above it. The 95% Clopper-Pearson confidence interval [0.0%, 100.0%] reflects the extreme non-significance: the data are fully compatible with any rate ratio between near-zero and infinity.

4.2 Magnitude Sweep (Dose-Response)

If tidal triggering exists and strengthens with magnitude (as Ide et al. predicted), the rate ratio should increase monotonically from M7.0+ to M8.5+. The results show no such pattern:

Threshold	N events	Syzygy hits	Rate ratio	p-value	Interpretation
M7.0+	390	161 (41.3%)	1.02×	0.4163	Near-null
M7.5+	**147**	**52 (35.4%)**	**0.87×**	**0.9175**	**Pre-registered primary**
M8.0+	40	19 (47.5%)	1.17×	0.2336	Non-significant
M8.5+	8	4 (50.0%)	1.23×	0.4207	N too small (8 events)

The dose-response pattern is incoherent: the rate ratio dips below 1.0 at M7.5+ before recovering to slightly above 1.0 at higher magnitudes. The sequence 1.02 → 0.87 → 1.17 → 1.23 shows no monotonic structure. The slight elevation at M8.0+ (RR=1.17×) and M8.5+ (RR=1.23×) is not significant at any threshold, and the M8.5+ bin contains only 8 events — statistically indistinguishable from chance.

The Bonferroni-corrected threshold for the magnitude sweep is p = 0.0025. The most favorable result (M8.0+, p = 0.234) falls far short of this.

4.3 Window Sensitivity

The primary ±3 day window captures approximately 40.6% of calendar time. Testing narrower windows:

Window	Time fraction	Syzygy hits / N	Rate ratio	p-value
±1 day	13.6%	23 / 147	1.16×	0.2605
±2 days	27.1%	39 / 147	0.98×	0.5917
±3 days (primary)	**40.6%**	**52 / 147**	**0.87×**	**0.9175**

The ±1 day window shows the most favorable rate ratio (1.16×, p = 0.26). This is the only configuration with a rate ratio above 1.0, and it is the narrowest window, capturing events within 24 hours of exact syzygy. This represents a possible — but far from conclusive — signal; a rate 16% above baseline with p = 0.26 cannot be distinguished from noise at reasonable sample sizes. If a tidal signal exists at all, it may be concentrated very close to syzygy, but confirming this would require substantially more events.

4.4 Robustness Checks

Check	N events	Rate ratio	p-value	Consistent with null?
Standard declustering (1.0×) — primary	147	0.87×	0.9175	Yes
Extended declustering (1.5×)	128	0.90×	0.8395	Yes
All tectonic settings (no zone filter)	240	1.00×	0.4997	Yes (exactly null)
Permutation test (10,000 shuffles)	147	—	0.9127	Yes
Pre-1990 temporal half	52	0.90×	0.7698	Yes
Post-1990 temporal half	95	0.85×	0.8997	Yes

Every robustness check returns a null result. The rate ratio ranges from 0.87× to 1.17× across all configurations, with all p-values far above conventional thresholds. The temporal split is particularly informative: both the pre-1990 and post-1990 halves independently yield null results, ruling out a period-specific effect driven by changes in seismic network coverage.

The all-settings test (RR = 1.00×, p = 0.4997) is a near-perfect null: non-subduction earthquakes show exactly the null expectation, and their inclusion does not affect the conclusion. The tectonic setting restriction (subduction zone only) was scientifically motivated by ocean-loading theory but makes no practical difference to the result.

4.5 Syzygy vs. Quadrature

A direct comparison of syzygy and quadrature periods provides the starkest summary of the result:

- Syzygy events (within ±3 days of new or full moon): **52** (new moon: 27, full moon: 25)
- Quadrature events (within ±3 days of first or last quarter): **54**

- Syzygy/Quadrature ratio: **0.96** (essentially 1:1)

If tides trigger earthquakes, syzygy events should substantially outnumber quadrature events. The observed ratio of 0.96 means syzygy and quadrature are **statistically interchangeable** as predictors of M7.5+ earthquake occurrence. The 1:1 ratio is precisely what the null hypothesis predicts — though note that quadrature windows also occupy ~40% of time, so this comparison is approximately equal-time.

4.6 Zone Breakdown

Syzygy rates by tectonic zone (M7.5+, ±3 days):

Zone	N events	Syzygy rate	Note
Chile-Peru	31	48.4%	Slightly above null; largest zone
Tonga-Kermadec	15	40.0%	Exactly at null expectation
Solomon Islands	15	20.0%	Below null — opposite of predicted
Central America	13	30.8%	Below null
Japan Trench	12	41.7%	Near null
Sumatra-Java	11	45.5%	Slightly above null
Kuril-Kamchatka	11	45.5%	Slightly above null
Alaska-Aleutian	8	50.0%	Above null; N too small to interpret

Syzygy rates across zones range from 16.7% (Izu-Bonin-Mariana) to 50.0% (Alaska-Aleutian, n=8). There is no consistent pattern: some zones are above the null expectation, others below. This scatter is consistent with random sampling variation around the null rate of 40.6%. If tidal triggering were setting-specific — for example, affecting only shallow high-loading zones like Sumatra or Chile — we would expect those zones to stand out. They do not.

5. Discussion

5.1 On the Ide et al. 2016 Result

Our analysis strongly suggests that the Ide et al. (2016) finding is a **statistical artifact** rather than a physical signal. Three lines of evidence support this conclusion:

1. Non-replication across all magnitude bins. Ide et al. found the strongest effect at M≥8.2. Our M8.0+ bin (n=40, RR=1.17×, p=0.234) and M8.5+ bin (n=8, RR=1.23×, p=0.421) show slight positive trends but are nowhere near significant. The M8.5+ result especially — the bin most comparable to Ide et al.'s M≥8.2 — is based on 8 events, and 4 in syzygy is one coin flip away from the null expectation of 3.25.

2. No dose-response pattern. The core prediction of the Ide et al. cascade model is that tidal influence should increase monotonically with magnitude. The sequence 1.02× → 0.87× → 1.17× → 1.23× is not monotonic. The dip at M7.5+ followed by recovery at M8.0+ is consistent with random fluctuation, not physical mechanism.

3. Small-N vulnerability. Ide et al.'s M≥8.2 bin contained approximately 12 events spanning multiple decades. At that sample size, the probability of a spurious $p < 0.05$ result even under a true null is substantial. Our study, with 147 primary events and 65 years of data, provides 12× the statistical power for comparable magnitude thresholds.

5.2 Physical Implications

The null result does not rule out all tidal influence on seismicity. Several physical scenarios remain consistent with our findings:

Tidal triggering of microseismicity. Multiple studies have demonstrated statistically significant tidal triggering of M1–3 earthquakes in specific tectonic settings, particularly at mid-ocean ridges and volcanic systems. This is not inconsistent with a null result for M7.5+ events: the triggering physics may operate at amplitudes too small to overcome the much higher stress thresholds for great earthquake nucleation.

Setting-specific effects below detection. Our zone breakdown is suggestive of heterogeneity — Chile-Peru (48.4%), Alaska-Aleutian (50.0%), and Sumatra-Java (45.5%) are all above null, while Solomon Islands (20.0%) and Central America (30.8%) are substantially below. A refined hypothesis targeting only the most ocean-loaded subduction zones might find a small positive signal, but at current catalog sizes this would be under-powered by construction.

The cascade model requires revision. Even if tidal stress occasionally initiates slow-slip events that in turn relieve stress gradually, such cascades would not necessarily manifest as a temporal clustering of mainshocks near syzygy. The timescale mismatch — tidal cycles of 14 days versus slow-slip events lasting weeks to months — makes direct temporal correlation unlikely.

5.3 The Three Largest Earthquakes

The three greatest recorded earthquakes all fall within syzygy windows, which may appear striking:

- M9.5 1960 Chilean earthquake: phase = 332° (new moon, 2.3 days before)
- M9.2 1964 Alaska earthquake: phase = 185° (full moon, 0.3 days after)

- M9.1 2004 Sumatra-Andaman earthquake: phase = 174° (full moon, 0.5 days before)

However, this requires no explanation beyond chance. The null hypothesis predicts that roughly 40% of all great earthquakes — including the very largest — will fall near syzygy. The 2004 Sumatra event is particularly close to full moon, but the 2011 Tōhoku M9.0 earthquake — not in our syzygy hit list — occurred 5.3 days after full moon (in the non-syzygy window), and the 2010 Chile M8.8 occurred 3.2 days before full moon (just at the syzygy boundary). The pattern is consistent with random sampling.

5.4 Limitations

- **Catalog completeness pre-1970:** Before the WWSSN was fully operational, smaller events in the M7.0–7.3 range may be underrepresented. This affects the lower magnitude bins more than the primary M7.5+ test.
- **Tidal metric choice:** We used a binary phase-based classification (syzygy vs. non-syzygy). A continuous tidal stress amplitude metric — accounting for perigee-apogee variations in lunar distance and Earth-Sun distance variations — might be slightly more powerful but would not change the fundamental null result given the magnitude of the effect size absent.
- **Slow-slip as intermediate step:** If tidal triggering operates through slow-slip intermediates, the relevant timescale may be weeks rather than days, making the ±3 day window too narrow. Testing ±2-week windows would require a different null expectation and was not pre-registered.
- **Spatial resolution:** Our subduction zone polygons use 50 km buffers, which may include some intraplate events near trenches. Stricter spatial filtering would reduce N further.

6. Conclusion

> **The Syzygy Question, answered:** Do the Moon and Sun time great earthquakes? Based on 65 years of global seismicity, the answer is no. The rate of M7.5+ subduction zone earthquakes within ±3 days of new or full moon is 35.4% — below the null expectation of 40.6%. Every robustness check, every magnitude threshold, every time period, and every tectonic zone tested is consistent with the null hypothesis. Tidal triggering of great earthquakes is not supported by the available data.

This finding has three implications:

For the Ide et al. controversy: Our pre-registered, well-powered replication finds no evidence for the cascade tidal triggering mechanism. The original result is most likely a statistical artifact of small sample sizes and post-hoc analysis. The Nature Geoscience paper should be interpreted with caution, particularly its higher-magnitude claims which rest on approximately 10 events.

For earthquake forecasting: Tidal phase is not a useful predictor of great earthquake timing. Syzygy and quadrature are interchangeable. Operational earthquake probability forecasts should not assign elevated probability to syzygy windows.

For future research: The most promising direction is a *setting-specific* analysis targeting the highest ocean-loading environments (Cascadia, Hikurangi, Nankai) with a continuous tidal stress metric rather than a binary phase classification. If any tidal signal exists for great earthquakes, it is small enough (RR < 1.2×) to require a dataset substantially larger than the 65-year USGS catalog to detect at conventional significance.

This study was conducted as part of the Earth Systems Observatory Thesis B investigation. All methods were pre-registered, the full backfill dataset is archived in `thesis-b-backfill.json`, and the analysis code is in `backfill-thesis-b.py`. The ESO real-time panel continues to monitor prospective M7.5+ subduction events against syzygy windows as new data accumulate.

7. References

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Appendix: Selected Syzygy Hit Events (M8.0+)

The following M8.0+ events fell within syzygy windows (±3 days of new or full moon). Their presence in this list does not indicate tidal causation; at a null rate of 40.6%, we expect approximately 40% of all great earthquakes to appear here.

Mag	Date / Time (UTC)	Phase	Moon	Zone	Event
M9.5	1960-05-22 19:11	332°	New	Chile-Peru	1960 Great Chilean (Valdivia)
M9.2	1964-03-28 03:36	185°	Full	Alaska-Aleutian	1964 Prince William Sound
M9.1	2004-12-26 00:58	174°	Full	Sumatra-Java	2004 Sumatra-Andaman
M8.8	2010-02-27 06:34	164°	Full	Chile-Peru	2010 Maule, Chile
M8.4	2001-06-23 20:33	29°	New	Chile-Peru	Atico, Peru
M8.4	2007-09-12 11:10	10°	New	Sumatra-Java	Bengkulu, Indonesia
M8.3	1994-10-04 13:22	349°	New	Kuril-Kamchatka	Shikotan, Russia
M8.3	2013-05-24 05:44	168°	Full	Kuril-Kamchatka	2013 Sea of Okhotsk
M8.2	1969-08-11 21:27	343°	New	Japan Trench	1969 Hokkaido Toho-oki
M8.2	2003-09-25 19:50	352°	New	Japan Trench	2003 Tokachi-Oki
M8.2	2014-04-01 23:46	31°	New	Chile-Peru	Iquique, Chile
M8.2	2017-09-08 04:49	199°	Full	Central America	2017 Tehuantepec, Mexico
M8.1	2007-04-01 20:39	173°	Full	Solomon Islands	2007 Solomon Islands
M8.0	1976-01-14 16:47	148°	Full	Tonga-Kermadec	1976 Kermadec Islands
M8.0	1986-05-07 22:47	352°	New	Alaska-Aleutian	1986 Atka, Alaska
M8.0	1995-07-30 05:11	27°	New	Chile-Peru	Antofagasta, Chile
M8.0	1995-10-09 15:35	192°	Full	Central America	1995 Colima-Jalisco, Mexico
M8.0	2007-08-15 23:40	35°	New	Chile-Peru	San Vicente, Peru

Note: M7.5-7.9 events omitted from this appendix for space. The complete hit event list (52 events) is available in thesis-b-backfill.json and the ESO Thesis B panel.